

Supporting Document

Mandatory Technical Document



PP-Module for LiFi Access Systems

Version: 1.0

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National Information Assurance Partnership

Foreword

This is a Supporting Document (SD), intended to complement the Common Criteria version 3 and the associated Common Evaluation Methodology for Information Technology Security Evaluation.

SDs may be “Guidance Documents”, that highlight specific approaches and application of the standard to areas where no mutual recognition of its application is required, and as such, are not of normative nature, or “Mandatory Technical Documents”, whose application is mandatory for evaluations whose scope is covered by that of the SD. The usage of the latter class is not only mandatory, but certificates issued as a result of their application are recognized under the CCRA.

Technical Editor:

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Version	Date	Comment
1.0	2026-02-24	Initial Release

General Purpose:

The purpose of this SD is to define evaluation methods for the functional behavior of LiFi Access System products.

Acknowledgments:

This SD was developed with support from NIAP LiFi Access Systems Technical Community members, with

representatives from industry, government agencies, Common Criteria Test Laboratories, and members of academia.

Table of Contents

- 1 Introduction
 - 1.1 Technology Area and Scope of Supporting Document
 - 1.2 Structure of the Document
 - 1.3 Terms
 - 1.3.1 Common Criteria Terms
 - 1.3.2 Technical Terms
- 2 Evaluation Activities for SFRs
 - 2.1 Network Device
 - 2.1.1 Modified SFRs
 - 2.2 TOE SFR Evaluation Activities
 - 2.2.1 Security Audit (FAU)
 - 2.2.2 Cryptographic Support (FCS)
 - 2.2.3 Identification and Authentication (FIA)
 - 2.2.4 Security Management (FMT)
 - 2.2.5 Protection of the TSF (FPT)
 - 2.2.6 TOE Access (FTA)
 - 2.2.7 Trusted Path/Channels (FTP)
 - 2.3 Evaluation Activities for Optional SFRs
 - 2.4 Evaluation Activities for Selection-Based SFRs
 - 2.4.1 Cryptographic Support (FCS)
 - 2.5 Evaluation Activities for Objective SFRs
 - 2.6 Evaluation Activities for Implementation-dependent SFRs
 - 2.6.1 SFRs for 802.11bb Implementations
 - 2.6.2 SFRs for Distributed TOE Implementations
 - 2.6.3 SFRs for G.vlc Implementations
- 3 Evaluation Activities for SARs
- 4 Required Supplementary Information
- Appendix A - References

1 Introduction

1.1 Technology Area and Scope of Supporting Document

The scope of the PP-Module for LiFi Access Systems is to describe the security functionality of LiFi Access Systems products in terms of [CC] and to define functional and assurance requirements for them. The PP-Module is intended for use with the following Base-PP:

- Network Device, version 4.0.

This SD is mandatory for evaluations of TOEs that claim conformance to a PP-Configuration that includes the PP-Module for :

- LiFi Access Systems, Version 1.0

As such it defines Evaluation Activities for the functionality described in the PP-Module as well as any impacts to the Evaluation Activities to the Base-PP(s) it modifies.

Although Evaluation Activities are defined mainly for the evaluators to follow, in general they also help developers to prepare for evaluation by identifying specific requirements for their TOE. The specific requirements in Evaluation Activities may in some cases clarify the meaning of Security Functional Requirements (SFR), and may identify particular requirements for the content of Security Targets (ST) (especially the TOE Summary Specification), user guidance documentation, and possibly supplementary information (e.g. for entropy analysis or cryptographic key management architecture).

1.2 Structure of the Document

Evaluation Activities can be defined for both SFRs and Security Assurance Requirements (SAR), which are themselves defined in separate sections of the SD.

If any Evaluation Activity cannot be successfully completed in an evaluation, then the overall verdict for the evaluation is a 'fail'. In rare cases there may be acceptable reasons why an Evaluation Activity may be modified or deemed not applicable for a particular TOE, but this must be approved by the Certification Body for the evaluation.

In general, if all Evaluation Activities (for both SFRs and SARs) are successfully completed in an evaluation then it would be expected that the overall verdict for the evaluation is a 'pass'. To reach a 'fail' verdict when the Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

Similarly, at the more granular level of assurance components, if the Evaluation Activities for an assurance component and all of its related SFR Evaluation Activities are successfully completed in an evaluation then it would be expected that the verdict for the assurance component is a 'pass'. To reach a 'fail' verdict for the assurance component when these Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

1.3 Terms

The following sections list Common Criteria and technology terms used in this document.

1.3.1 Common Criteria Terms

Assurance	Grounds for confidence that a TOE meets the SFRs [CC].
Base Protection Profile (Base-PP)	Protection Profile used as a basis to build a PP-Configuration.
Collaborative Protection Profile (cPP)	A Protection Profile developed by international technical communities and approved by multiple schemes.
Common Criteria (CC)	Common Criteria for Information Technology Security Evaluation (International Standard ISO/IEC 15408).
Common Criteria Testing Laboratory	Within the context of the Common Criteria Evaluation and Validation Scheme (CCEVS), an IT security evaluation facility accredited by the National Voluntary Laboratory Accreditation Program (NVLAP) and approved by the NIAP Validation Body to conduct Common Criteria-based evaluations.

Common Evaluation Methodology (CEM)	Common Evaluation Methodology for Information Technology Security Evaluation.
Distributed TOE	A TOE composed of multiple components operating as a logical whole.
Operational Environment (OE)	Hardware and software that are outside the TOE boundary that support the TOE functionality and security policy.
Protection Profile (PP)	An implementation-independent set of security requirements for a category of products.
Protection Profile Configuration (PP-Configuration)	A comprehensive set of security requirements for a product type that consists of at least one Base-PP and at least one PP-Module.
Protection Profile Module (PP-Module)	An implementation-independent statement of security needs for a TOE type complementary to one or more Base-PPs.
Security Assurance Requirement (SAR)	A requirement to assure the security of the TOE.
Security Functional Requirement (SFR)	A requirement for security enforcement by the TOE.
Security Target (ST)	A set of implementation-dependent security requirements for a specific product.
Target of Evaluation (TOE)	The product under evaluation.
TOE Security Functionality (TSF)	The security functionality of the product under evaluation.
TOE Summary Specification (TSS)	A description of how a TOE satisfies the SFRs in an ST.

1.3.2 Technical Terms

Access Point (AP)	A device that provides the network interface that enables wireless client hosts to access a wired network.
LiFi	A wireless computer network that links two or more devices using modulated optical communication to form a local area network within a limited area such as a home, school, computer laboratory, campus, office building, etc.
Service Set Identifier (SSID)	The primary name associated with an 802.11 wireless local area network (WLAN).

2 Evaluation Activities for SFRs

The EAs presented in this section capture the actions the evaluator performs to address technology specific aspects covering specific SARs (e.g. ASE_TSS.1, ADV_FSP.1, AGD_OPE.1, and ATE_IND.1) – this is in addition to the

CEM workunits that are performed in Section [3 Evaluation Activities for SARs](#).

Regarding design descriptions (designated by the subsections labeled TSS, as well as any required supplementary material that may be treated as proprietary), the evaluator must ensure there is specific information that satisfies the EA. For findings regarding the TSS section, the evaluator's verdicts will be associated with the CEM workunit ASE_TSS.1-1. Evaluator verdicts associated with the supplementary evidence will also be associated with ASE_TSS.1-1, since the requirement to provide such evidence is specified in ASE in the PP.

For ensuring the guidance documentation provides sufficient information for the administrators/users as it pertains to SFRs, the evaluator's verdicts will be associated with CEM workunits ADV_FSP.1-7, AGD_OPE.1-4, and AGD_OPE.1-5.

Finally, the subsection labeled Tests is where the authors have determined that testing of the product in the context of the associated SFR is necessary. While the evaluator is expected to develop tests, there may be instances where it is more practical for the developer to construct tests, or where the developer may have existing tests. Therefore, it is acceptable for the evaluator to witness developer-generated tests in lieu of executing the tests. In this case, the evaluator must ensure the developer's tests are executing both in the manner declared by the developer and as mandated by the EA. The CEM workunits that are associated with the EAs specified in this section are: ATE_IND.1-3, ATE_IND.1-4, ATE_IND.1-5, ATE_IND.1-6, and ATE_IND.1-7.

2.1 Network Device

The EAs defined in this section are only applicable in cases where the TOE claims conformance to a PP-Configuration that includes the NDcPP.

2.1.1 Modified SFRs

The PP-Module does not modify any requirements when the NDcPP is the base.

2.2 TOE SFR Evaluation Activities

2.2.1 Security Audit (FAU)

FAU_GEN.1/LiFi Audit Data Generation (LiFi)

FAU_GEN.1/LiFi

TSS

There are no additional TSS evaluation activities for this component.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall complete the evaluation activity for FAU_GEN.1 as described in the NDcPP for the auditable events defined in the PP-Module, in addition to the applicable auditable events that are defined in the NDcPP. The evaluator shall also ensure that the administrative actions defined for this PP-Module are appropriately audited.

2.2.2 Cryptographic Support (FCS)

FCS_COP.1/LiFiDataEncryption Cryptographic Operation (Data Encryption and Decryption for LiFi)

FCS_COP.1/LiFiDataEncryption

TSS

The evaluator shall examine the TSS to verify that it describes how the TOE generates a key based on output from a random bit generator as specified in FCS_RBG.1. The evaluator shall review the TSS to verify that it describes how the functionality described by FCS_RBG.1 is invoked.

The evaluator shall examine the TSS to verify that it identifies the usage and key lifecycle for keys generated using each selected algorithm.

Guidance

The evaluator shall verify that the guidance instructs the administrator how to configure the TOE to generate keys for the selected key generation algorithms for all key types and uses identified in the TSS.

Tests

If the TOE claims support for AES-CBC, AES-GCM, or AES-GCMP, the evaluator shall perform the testing defined in the Base-PP's FCS_COP.1/SKC SFR for the applicable modes.

AES-CCM Tests

The evaluator shall test the generation-encryption and decryption-verification functionality of AES-CCM for the following input parameter and tag lengths:

128 bit and 256 bit keys

Two payload lengths. One payload length will be the shortest supported payload length, greater than or equal to zero bytes. The other payload length will be the longest supported payload length, less than or equal to 32 bytes (256 bits).

Two or three associated data lengths. One associated data length will be 0, if supported. One associated data length will be the shortest supported payload length, greater than or equal to zero bytes. One associated data length will be the longest supported payload length, less than or equal to 32 bytes (256 bits). If the implementation supports an associated data length of 216 bytes, an associated data length of 216 bytes will be tested.

Nonce lengths. All supported nonce lengths between 7 and 13 bytes, inclusive, will be tested.

Tag lengths. All supported tag lengths of 4, 6, 8, 10, 12, 14, and 16 bytes will be tested.

Due to the restrictions that IEEE 802.11 specifies for this mode (nonce length of 13 and tag length of 8), it is acceptable to test a subset of the supported lengths as long as the selections fall into the ranges specified above. In this case, the evaluator shall ensure that these are the only supported lengths. To test the generation-encryption functionality of AES-CCM, the evaluator shall perform the following four tests:

- Test FCS_COP.1/LiFiDataEncryption:1: For each supported key and associated data length and any supported payload, nonce, and tag length, the evaluator shall supply one key value, one nonce value, and 10 pairs of associated data and payload values, and obtain the resulting ciphertext.
- Test FCS_COP.1/LiFiDataEncryption:2: For each supported key and payload length and any supported associated data, nonce, and tag length, the evaluator shall supply one key value, one nonce value, and 10 pairs of associated data and payload values, and obtain the resulting ciphertext.
- Test FCS_COP.1/LiFiDataEncryption:3: For each supported key and nonce length and any supported associated data, payload, and tag length, the evaluator shall supply one key value and 10 associated data, payload, and nonce value 3-tuples, and obtain the resulting ciphertext.
- Test FCS_COP.1/LiFiDataEncryption:4: For each supported key and tag length and any supported associated data, payload, and nonce length, the evaluator shall supply one key value, one nonce value, and 10 pairs of associated data and payload values and, obtain the resulting ciphertext.

To determine correctness in each of the above tests, the evaluator shall compare the ciphertext with the result of generation-encryption of the same inputs with a known good implementation.

To test the decryption-verification functionality of AES-CCM, for each combination of supported associated data length, payload length, nonce length, and tag length, the evaluator shall supply a key value and 15 nonce, associated data, and ciphertext 3-tuples, and obtain either a fail result or a pass result with the decrypted payload.

The evaluator shall supply 10 tuples that should fail and 5 that should pass per set of 15.

Additionally, the evaluator shall use tests from the IEEE 802.11-02/362r6 document “Proposed Test vectors for IEEE 802.11 TGi,” dated September 10, 2002, Section 2.1 AES-CCMP Encapsulation Example and Section 2.2 Additional AES-CCMP Test Vectors to verify further the IEEE 802.11-2020 implementation of AES-CCMP.

2.2.3 Identification and Authentication (FIA)

FIA_8021X_EXT.1 802.1X Authentication

FIA_8021X_EXT.1

TSS

In order to show that the TSF implements the 802.1X-2010 standard correctly, the evaluator shall ensure that the TSS contains the following information:

- The sections (clauses) of the standard that the TOE implements
- For each identified section, any options selected in the implementation allowed by the standards are specified
- For each identified section, any non-conformance is identified and described, including a justification for the non-conformance

Because the connection to the RADIUS server will be contained in an IPsec or RadSec (TLS) tunnel, the security mechanisms detailed in the RFCs identified in the requirement are not relied on to provide protection for these communications. Consequently, no extensive analysis of the RFCs is required. However, the evaluator shall ensure that the TSS describes the measures (documentation, testing) that are taken by the product developer to ensure that the TOE conforms to the RFCs listed in this requirement.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall perform the following tests:

- Test FIA_8021X_EXT.1:1: The evaluator shall demonstrate that a wireless client has no access to the test network. After successfully authenticating with a RADIUS server through the TOE, the evaluator shall demonstrate that the wireless client does have access to the test network.
- Test FIA_8021X_EXT.1:2: The evaluator shall demonstrate that a wireless client has no access to the test network. The evaluator shall attempt to authenticate using an invalid client certificate, such that the EAP-TLS negotiation fails. This should result in the wireless client still being unable to access the test network.
- Test FIA_8021X_EXT.1:3: The evaluator shall demonstrate that a wireless client has no access to the test network. The evaluator shall attempt to authenticate using an invalid RADIUS certificate, such that the EAP-TLS negotiation fails. This should result in the wireless client still being unable to access the test network.

Note: Tests 2 and 3 above are not tests that "EAP-TLS works," although that is a by-product of the test. The test is actually that a failed authentication (under two failure modes) results in denial of access to the network, which demonstrates the enforcement of FIA_8021X_EXT.1.3.

FIA_UAU.6 Re-authenticating

FIA_UAU.6

TSS

There are no additional TSS evaluation activities for this component.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall attempt to change their password as directed by the operational guidance. While making this attempt, the evaluator shall verify that re-authentication is required.

If other re-authentication conditions are specified, the evaluator shall cause those conditions to occur and verify that the TSF re-authenticates the authenticated user.

2.2.4 Security Management (FMT)

FMT_SMF.1/LiFi Specification of Management Functions

FMT_SMF.1/LiFi

TSS

The evaluator shall examine the TSS to verify that it describes the security policy configuration in terms of the attributes claimed in the SFR.

The evaluator shall also confirm that the TSS includes a description of how connection attempts from clients that are not operating on an approved security type are handled.

Guidance

The evaluator shall examine the operational guidance to verify that it includes procedures for configuring the security policy of the wireless network including all attributes claimed in the ST.

Tests

The evaluator shall perform the following tests:

- Test FMT_SMF.1/LiFi:1: For each security type specified in the TSS, configure the network to the approved security type and verify that the client can establish a connection.

Maintaining the same SSID, change the security type of the client to a non-approved security type and attempt to establish a connection. Verify that the connection was unsuccessful.

- Test FMT_SMF.1/LiFi:2: For each authentication protocol specified in the TSS, configure the network accordingly per the AGD. Verify that the client connection attempt is successful when using the correct client credentials and that the connection is unsuccessful when incorrect authentication credentials are used.
- Test FMT_SMF.1/LiFi:3: [Conditional: SSID is broadcast] The evaluator shall configure the SSID to be broadcast. Using a network sniffing tool, the evaluator shall capture a beacon frame and confirm that the SSID is included. The evaluator shall then configure the SSID to be hidden. Using a network sniffing tool, the evaluator shall capture a beacon frame and confirm that the SSID is not listed.

FMT_SMR_EXT.1 No Administration from Client

FMT_SMR_EXT.1

TSS

There are no additional TSS evaluation activities for this component.

Guidance

The evaluator shall review the operational guidance to ensure that it contains instructions for administering the TOE both locally and remotely, including any configuration that needs to be performed on the client for remote administration. The evaluator shall confirm that the TOE does not permit remote administration from a wireless client by default.

Tests

The evaluator shall demonstrate that after configuring the TOE for first use from the operational guidance, it is possible to establish an administrative session with the TOE on the “wired” portion of the device. The evaluator

shall then demonstrate that an identically configured wireless client that can successfully connect to the TOE cannot be used to perform administration.

2.2.5 Protection of the TSF (FPT)

FPT_ACF_EXT.1 Access Controls

FPT_ACF_EXT.1.1

TSS

The evaluator shall confirm that the TSS specifies the locations of kernel drivers and modules, security audit logs, shared libraries, system executables, and system configuration files. Every file does not need to be individually identified, but the system's conventions for storing and protecting such files must be specified.

Guidance

There are no additional Guidance evaluation activities for this element.

Tests

For these tests the evaluator shall use appropriate development environment (e.g., a Virtual Machine) and development tools (debuggers, simulators, etc.), including enabling root shell access as needed. The evaluator shall create an unprivileged user account. Using this account, the evaluator shall ensure that the following tests result in a negative outcome (i.e., the action results in the OS denying the evaluator permission to complete the action):

- Test FPT_ACF_EXT.1.1:1: The evaluator shall attempt to modify all kernel drivers and modules.
- Test FPT_ACF_EXT.1.1:2: The evaluator shall attempt to modify all security audit logs generated by the logging subsystem.
- Test FPT_ACF_EXT.1.1:3: The evaluator shall attempt to modify all shared libraries that are used throughout the system.
- Test FPT_ACF_EXT.1.1:4: The evaluator shall attempt to modify all system executables.
- Test FPT_ACF_EXT.1.1:5: The evaluator shall attempt to modify all system configuration files.
- Test FPT_ACF_EXT.1.1:6: The evaluator shall attempt to modify any additional components selected.

FPT_ACF_EXT.1.2

TSS

There are no additional TSS evaluation activities for this element.

Guidance

There are no additional Guidance evaluation activities for this element.

Tests

For these tests the evaluator shall use appropriate development environment (e.g., a Virtual Machine) and development tools (debuggers, simulators, etc.), including enabling root shell access as needed. The evaluator shall create an unprivileged user account. Using this account, the evaluator shall ensure that the following tests result in a negative outcome (i.e., the action results in the OS denying the evaluator permission to complete the action):

- Test FPT_ACF_EXT.1.2:1: The evaluator shall attempt to read security audit logs generated by the auditing subsystem.
- Test FPT_ACF_EXT.1.2:2: The evaluator shall attempt to read system-wide credential repositories.
- Test FPT_ACF_EXT.1.2:3: The evaluator shall attempt to read any other object specified in the assignment.

FPT_AS LR_EXT.1 Address Space Layout Randomization

FPT_AS LR_EXT.1

TSS

The evaluator shall examine the TSS to verify that it identifies the number of entropy bits used for ASLR and what exceptions there are to this, if any.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

For these tests the evaluator shall use appropriate development environment (e.g., a Virtual Machine) and development tools (debuggers, simulators, etc.), including enabling root shell access as needed. The evaluator shall select three executables included with the TSF. If the TSF includes a web server it must be selected. The evaluator shall prioritize executables that are UDP or TCP listening network services. For each of these executables, the evaluator shall launch the same executable twice and inspect the memory map using an appropriate tool or interface (e.g. /proc/\$PID/maps on a Linux-based OS). The evaluator shall ensure that no memory mappings are placed in the same location. If the rare chance occurs that two mappings are the same for a single executable and not the same for another chosen executable, the evaluator shall repeat the test with that executable to verify that in the second test the mappings are different. This test should also be completed on the same hardware. The evaluator shall also inspect the executables to verify the fPIE compile flag is enabled. The evaluator shall read /proc/kallsyms and note the _text address. The evaluator shall then completely reboot the TSF and re-read /proc/kallsyms and in verify that the addresses differ. In the rare event KASLR is on and the addresses are the same the evaluator shall reboot and verify again.

FPT_SBOP_EXT.1 Stack Buffer Overflow Protection

FPT_SBOP_EXT.1

TSS

The evaluator shall examine the TSS to verify that it describes how stack-based buffer overflow protection is implemented, or how the TOE is designed in such a way that these protections are not explicitly needed.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

For stack-based OSes, the evaluator shall determine that the TSS contains a description of stack-based buffer overflow protections used by the OS. These are referred to by a variety of terms. These include, but are not limited to, tagging, stack cookie, stack guard, and stack canaries. The TSS must include a rationale for any binaries that are not protected in this manner. The evaluator shall also preform the following test. For this test the evaluator shall use an appropriate development environment (e.g., a Virtual Machine) and development tools (debuggers, simulators, etc.), including enabling root shell access as needed.

- Test FPT_SBOP_EXT.1:1: The evaluator shall inventory the kernel, libraries, and application binaries to determine those that do not implement stack-based buffer overflow protections. This list should match up with the list provided in the TSS.

For OSes that store parameters or variables separately from control flow values, the evaluator shall verify that the TSS describes what data structures control values, parameters, and variables are stored. The evaluator shall also ensure that the TSS includes a description of the safeguards that ensure parameters and variables do not intermix with control flow values.

FPT_W^X_EXT.1 Write XOR Execute Memory Pages

FPT_W^X_EXT.1

TSS

The evaluator shall inspect the vendor-provided developer documentation and verify that no memory-mapping can be made with write and execute permissions except for the cases listed in the assignment.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

- Test FPT_W^X_EXT.1:1: The evaluator shall acquire or construct a test program which attempts to allocate memory that is both writable and executable. The evaluator shall run the program and confirm that it fails to allocate memory that is both writable and executable.
- Test FPT_W^X_EXT.1:2: The evaluator shall acquire or construct a test program which allocates memory that is executable and then subsequently requests additional write or modify permissions on that memory. The evaluator shall run the program and confirm that at no time during the lifetime of the process is the memory both writable and executable.
- Test FPT_W^X_EXT.1:3: The evaluator shall acquire or construct a test program which allocates memory that is writable and then subsequently requests additional execute permissions on that memory. The evaluator shall run the program and confirm that at no time during the lifetime of the process is the memory both writable and executable.

2.2.6 TOE Access (FTA)

FTA_TSE.1 TOE Session Establishment

FTA_TSE.1

TSS

The evaluator shall examine the TSS to determine that all of the attributes on which a client session can be denied are specifically defined.

Guidance

The evaluator shall examine the operational guidance to determine that it contains guidance for configuring each of the attributes identified in the TSS.

Tests

For each supported attribute, the evaluator shall perform the following test:

- Test FTA_TSE.1:1: The evaluator shall successfully establish a client session with a wireless client. The evaluator shall then follow the operational guidance to configure the system so that the client's access is denied based on a specific value of the attribute. The evaluator shall then attempt to establish a session in contravention to the attribute setting (for instance, the client is denied network access based on the TOE interface that it is connecting to, or that the client is denied access based on the time-of-day or day-of-week it is attempting connection on). The evaluator shall observe that the access attempt fails.

2.2.7 Trusted Path/Channels (FTP)

FTP_ITC.1/8021X Inter-TSF Trusted Channel (802.1X)

FTP_ITC.1/8021X

The evaluator shall perform the following activities in addition to those required by the NDcPP:

TSS

The evaluator shall examine the TSS to determine that for all communications with authorized IT entities identified in the requirement, each communications mechanism is identified in terms of the allowed protocols for that IT entity. The evaluator shall also confirm that all protocols listed in the TSS are specified and included in the requirements in the ST.

Guidance

The evaluator shall confirm that the guidance documentation contains instructions for establishing the allowed protocols with each authorized IT entity and that it contains recovery instructions should a connection be unintentionally broken.

Tests

The evaluator shall perform the following tests:

- Test FTP_ITC.1/8021X:1: The evaluator shall ensure that communications using each protocol with each authorized IT entity is tested during the course of the evaluation, setting up the connections as described in the guidance documentation and ensuring that communication is successful.
- Test FTP_ITC.1/8021X:2: For each protocol that the TOE can initiate as defined in the requirement, the evaluator shall follow the guidance documentation to ensure that the communication channel can be initiated from the TOE.
- Test FTP_ITC.1/8021X:3: The evaluator shall ensure, for each communication channel with an authorized IT entity, the channel data is not sent in plaintext.
- Test FTP_ITC.1/8021X:4: The evaluator shall, for each protocol associated with each authorized IT entity tested during test 1, physically interrupt an established connection. The evaluator shall ensure that when physical connectivity is restored, communications are appropriately protected.

2.3 Evaluation Activities for Optional SFRs

The PP-Module does not define any optional requirements.

2.4 Evaluation Activities for Selection-Based SFRs

2.4.1 Cryptographic Support (FCS)

FCS_RADSEC_EXT.1 RadSec

FCS_RADSEC_EXT.1

TSS

The evaluator shall verify that the TSS description includes the use of RADIUS over TLS, as described in RFC 6614. If X.509v3 certificates is selected, the evaluator shall ensure that the TSS description includes the use of client-side certificates for TLS mutual authentication.

Guidance

The evaluator shall verify that any configuration necessary to meet the requirement must be contained in the guidance.

Tests

The evaluator shall demonstrate the ability to successfully establish a RADIUS over TLS connection with a RADIUS server. This test will be performed with X.509v3 certificates if selected and performed with pre-shared keys if selected.

FCS_RADSEC_EXT.2 RadSec Using Pre-Shared Keys

FCS_RADSEC_EXT.2

TSS

The evaluator shall check the description of the implementation of this protocol in the TSS to ensure that the ciphersuites supported are specified. The evaluator shall check the TSS to ensure that the ciphersuites specified are identical to those listed for this component. The evaluator shall also verify that the TSS contains a description of the denial of old SSL and TLS versions. The evaluator shall examine the TSS to ensure it describes the process by which the bit-based pre-shared keys are generated (if the TOE supports this functionality), and confirm that this process uses the RBG specified in FCS_RBG.1.

Guidance

The evaluator shall verify that any configuration necessary to meet the requirement must be contained in the guidance.

The evaluator shall also check the guidance documentation to ensure that it contains instructions on configuring the TOE so that RADIUS over TLS conforms to the description in the TSS (for instance, the set of ciphersuites advertised by the TOE may have to be restricted to meet the requirements).

The evaluator shall confirm the operational guidance contains instructions for either entering bit-based pre-shared keys or generating a bit-based pre-shared key (or both).

Tests

The testing for this requirement is addressed by the functional claims made in [Functional Package for Transport Layer Security \(TLS\), version 2.1](#) for this interface.

2.5 Evaluation Activities for Objective SFRs

The PP-Module does not define any objective requirements.

2.6 Evaluation Activities for Implementation-dependent SFRs

2.6.1 SFRs for 802.11bb Implementations

FCS_CKM.1/WPA Cryptographic Key Generation (Symmetric Keys for WPA2 Connections)

FCS_CKM.1/WPA

TSS

The cryptographic primitives will be verified through evaluation activities specified elsewhere in this PP-Module. The evaluator shall verify that the TSS describes how the primitives defined and implemented by this PP-Module are used by the TOE in establishing and maintaining secure connectivity to the wireless clients. This description will include how the GTK and PTK are generated or derived. The TSS shall also provide a description of the developer's methods of assuring that their implementation conforms to the cryptographic standards; this includes not only testing done by the developing organization, but also proof of third-party testing that is performed (e.g., WPA2 certification). The evaluator shall ensure that the description of the testing methodology is of sufficient detail to determine the extent to which the details of the protocol specifics are tested.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall perform the following test using a packet sniffing tool to collect frames between the TOE and a wireless client:

Step 1: The evaluator shall identify the channel used by the TOE and configure the sniffer to sniff only on that channel (i.e., lock the sniffer on the selected channel). The sniffer should also be configured to filter on the MAC address of the TOE and client.

Step 2: The evaluator shall configure the TOE to communicate with a LiFi client using IEEE 802.11-2020 and a 256-bit (64 hex values 0-f) pre-shared key, setting up the connections as described in the operational guidance. The pre-shared key is only used for testing.

Step 3: The evaluator shall start the sniffing tool, initiate a connection between the TOE and LiFi client, and allow the TOE to authenticate, associate, and successfully complete the four-way handshake with the client.

Step 4: The evaluator shall set a timer for one minute, at the end of which the evaluator shall disconnect the client from the TOE and stop the sniffer.

Step 5: The evaluator shall identify the four-way handshake frames (denoted EAPOL-key in Wireshark captures) and derive the PTK from the four-way handshake frames and pre-shared key as specified in IEEE 802.11-2020.

Step 6: The evaluator shall select the first data frame from the captured packets that was sent between the client and TOE after the four-way handshake successfully completed and without the frame control value 0x4208 (the first two bytes are 08 42). The evaluator shall use the PTK to decrypt the data portion of the packet as specified in IEEE 802.11-2020 and verify that the decrypted data contains ASCII-readable text.

Step 7: The evaluator shall repeat Step 6 for the next two data frames between the TOE and client, and without frame control value 0x4208.

Additionally, the evaluator shall test the PRF function using the test vectors from:

- Section 2.4 “The PRF Function – PRF (key, prefix, data, length)” of the IEEE 802.11-02/362r6 document “Proposed Test vectors for IEEE 802.11 TGi” dated September 10, 2002
- Annex J.3 “PRF reference implementation and test vectors” of IEEE 802.11-2020

FCS_CKM.2/GTK Cryptographic Key Distribution (GTK)

FCS_CKM.2/GTK

TSS

The evaluator shall check the TSS to ensure that it describes how the GTK is wrapped prior to being distributed using the AES implementation specified in this PP-Module, and also how the GTKs are distributed when multiple clients connect to the TOE.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall perform the following test using a packet sniffing tool to collect frames between a wireless client and the TOE (which may be performed in conjunction with the evaluation activity for FCS_CKM.1/PMK).

To fully test the broadcast and multicast functionality, these steps will be performed as the evaluator connects multiple clients to the TOE. The evaluator shall ensure that GTKs established are sent to the appropriate participating clients.

Step 1: The evaluator shall identify the channel used by the TOE and configure the sniffer to sniff only on that channel (i.e., lock the sniffer on the selected channel). The sniffer should also be configured to filter on the MAC address of the TOE and client.

Step 2: The evaluator shall configure the TOE to communicate with the client using IEEE 802.11-2020 and a 256-bit (64 hex values 0-f) pre-shared key, setting up the connections as described in the operational guidance. The pre-shared key is only used for testing.

Step 3: The evaluator shall start the sniffing tool, initiate a connection between the TOE and client, and allow the client to authenticate, associate, and successfully complete the four-way handshake with the TOE.

Step 4: The evaluator shall set a timer for one minute, at the end of which the evaluator shall disconnect the TOE from the client and stop the sniffer.

Step 5: The evaluator shall identify the four-way handshake frames (denoted EAPOL-key in Wireshark captures) and derive the PTK and GTK from the four-way handshake frames and pre-shared key as specified in IEEE 802.11-2020.

Step 6: The evaluator shall select the first data frame from the captured packets that was sent between the TOE and client after the four-way handshake successfully completed, and with the frame control value 0x4208 (the first two bytes are 08 42). The evaluator shall use the GTK to decrypt the data portion of the selected packet as specified in IEEE 802.11-2020 and verify that the decrypted data contains ASCII-readable text.

Step 7: The evaluator shall repeat Step 6 for the next two data frames with frame control value 0x4208.

The evaluator shall also perform the following testing based on the supported GTK distribution methods:

AES Key Wrap (AES-KW Tests)

- Test FCS_CKM.2/GTK:1: The evaluator shall test the authenticated encryption functionality of AES-KW for each combination of the following input parameter lengths:
 - 128 and 256 bit key encryption keys (KEKs)
 - Three plaintext lengths:
 1. One of the plaintext lengths will be two semi-blocks (128 bits).
 2. One of the plaintext lengths will be three semi-blocks (192 bits).
 3. The third data unit length will be the longest supported plaintext length less than or equal to 64 semi-blocks (4096 bits).

For each combination, generate a set of 100 key and plaintext pairs and obtain the ciphertext that results from AES-KW authenticated encryption. To determine correctness, the evaluator shall use the same key and plaintext values and encrypt them using a known good implementation of AES-KW authenticated-encryption, and ensure that the resulting respective ciphertext values are identical.

- Test FCS_CKM.2/GTK:2: The evaluator shall test the authenticated-decryption functionality of AES-KW using the same test as for authenticated-encryption, replacing plaintext values with ciphertext values and AES-KW authenticated-encryption with AES-KW authenticated-decryption. Additionally, the evaluator shall modify one byte of the ciphertext, attempt to decrypt the modified ciphertext, and ensure that a failure is returned rather than plaintext.

AES Key Wrap with Padding (AES-KWP Tests)

- Test FCS_CKM.2/GTK:3:

The evaluator shall test the authenticated-encryption functionality of AES-KWP for each combination of the following input parameter lengths:

128 and 256 bit key encryption keys (KEKs)

Three plaintext lengths. One plaintext length will be one octet. One plaintext length will be 20 octets (160 bits). One plaintext length will be the longest supported plaintext length less than or equal to 512 octets (4096 bits).

Use a set of 100 key and plaintext pairs and obtain the ciphertext that results from AES-KWP authenticated encryption. To determine correctness, the evaluator shall use the AES-KWP authenticated-encryption function of a known good implementation.

- Test FCS_CKM.2/GTK:4: The evaluator shall test the authenticated-decryption functionality of AES-KWP using the same test as for AES-KWP authenticated-encryption, replacing plaintext values with ciphertext values and AES-KWP authenticated-encryption with AES-KWP authenticated-decryption. Additionally, the evaluator shall modify one byte of the ciphertext, attempt to decrypt the modified ciphertext, and ensure that a failure is returned rather than plaintext.

FCS_CKM.2/PMK Cryptographic Key Distribution (PMK)

FCS_CKM.2/PMK

TSS

The evaluator shall examine the TSS to determine that it describes how the PMK is transferred (that is, through what EAP attribute) to the TOE.

Guidance

There are no additional Guidance evaluation activities for this component.

Tests

The evaluator shall establish a session between the TOE and a RADIUS server according to the configuration guidance provided. The evaluator shall then examine the traffic that passes between the RADIUS server and the TOE during a successful attempt to connect a wireless client to the TOE to determine that the PMK is not exposed.

FIA_PSK_EXT.1 Pre-Shared Key Composition

FIA_PSK_EXT.1

TSS

The evaluator shall verify that the TSS describes

1. the protocols that can use pre-shared keys and that these are consistent with the selections made in FIA_PSK_EXT.1.1.
2. the allowable values for pre-shared keys and that they are consistent with the selections made in FIA_PSK_EXT.1.2.
3. the way bit-based pre-shared keys are procured and that it is consistent with the selections made in FIA_PSK_EXT.1.3.

Guidance

The evaluator shall examine the operational guidance to determine that it provides guidance to administrators on the composition of strong text-based pre-shared keys, and (if the selection indicates keys of various lengths can be entered) that it provides information on the range of lengths supported. The guidance must specify the allowable characters for pre-shared keys, and that list must be a superset of the list contained in FIA_PSK_EXT.1.2.

The evaluator shall confirm the operational guidance contains instructions for either entering bit-based pre-shared keys for each protocol identified in the requirement or for generating a bit-based pre-shared key (or both).

Tests

The evaluator shall also perform the following tests for each protocol (or instantiation of a protocol, if performed by a different implementation on the TOE). Note that one or more of these tests can be performed with a single test case.

- Test FIA_PSK_EXT.1:1: The evaluator shall compose a pre-shared key of 22 characters that contains a combination of the allowed characters in accordance with the operational guidance and demonstrates that a successful protocol negotiation can be performed with the key.
- Test FIA_PSK_EXT.1:2: [conditional]: If the TOE supports pre-shared keys of multiple lengths, the evaluator shall repeat Test 1 using the minimum length; the maximum length; a length inside the allowable range; and invalid lengths beyond the supported range (both higher and lower). The minimum, maximum, and included length tests should be successful, and the invalid lengths must be rejected by the TOE.
- Test FIA_PSK_EXT.1:3: [conditional]: If the TOE does not generate bit-based pre-shared keys, the evaluator shall obtain a bit-based pre-shared key of the appropriate length and enter it according to the instructions in the operational guidance. The evaluator shall then demonstrate that a successful protocol negotiation can be performed with the key.
- Test FIA_PSK_EXT.1:4: [conditional]: If the TOE does generate bit-based pre-shared keys, the evaluator shall generate a bit-based pre-shared key of the appropriate length and use it according to the instructions in the operational guidance. The evaluator shall then demonstrate that a successful protocol negotiation can be performed with the key.

FTP_ITC.1/Client Inter-TSF Trusted Channel (LiFi 802.11 Client Access)

FTP_ITC.1/Client

This component is adequately evaluated when performing the evaluation activities for FTP_ITC.1 in the [Network Device, version 4.0](#) Base-PP. The evaluator shall repeat the tests from that SFR for each claimed LiFi communications channel.

2.6.2 SFRs for Distributed TOE Implementations

FCS_CKM.2/DISTRIB Cryptographic Key Distribution (802.11 Keys)

FCS_CKM.2/DISTRIB

TSS

The evaluator shall examine the TSS to determine that it describes which keys are distributed outside the TOE, where they are sent, and the purpose for this transfer.

Guidance

If this function is dependent on TOE configuration, the evaluator shall confirm that the operational guidance contains instructions for how to configure that the keys are adequately protected.

Tests

This requirement will be tested in conjunction with the tests for the cryptographic primitives, the secure protocols, and FPT_ITT.1 (Base-PP).

2.6.3 SFRs for G.vlc Implementations

FTP_ITC.1/Gvlc Inter-TSF Trusted Channel (MACsec)

FTP_ITC.1/Gvlc

G.vlc is implemented using MACsec. To demonstrate the implementation of G.vlc, the evaluator shall apply the test activities for FCS_MACSEC_EXT.1, FCS_MACSEC_EXT.2, FCS_MACSEC_EXT.3, and FCS_MACSEC_EXT.4 as defined in the PP-Module for MACsec Ethernet Encryption.

3 Evaluation Activities for SARs

The PP-Module does not define any SARs beyond those defined within the base NDcPP to which it must claim conformance. It is important to note that a TOE that is evaluated against the PP-Module is inherently evaluated against this Base-PP as well. The NDcPP includes a number of Evaluation Activities associated with both SFRs and SARs. Additionally, the PP-Module includes a number of SFR-based Evaluation Activities that similarly refine the SARs of the Base-PPs. The evaluation laboratory will evaluate the TOE against the Base-PP and supplement that evaluation with the necessary SFRs that are taken from the PP-Module.

4 Required Supplementary Information

This Supporting Document has no required supplementary information beyond the ST, operational guidance, and testing.

Appendix A - References

Identifier	Title
[NDcPP]	collaborative Protection Profile for Network Devices, Version 4.0, December 22, 2025